China Next-Year Life Expectancy Prediction Based on World Bank WDI Indicators

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Abstract

This project investigates China's next-year life expectancy prediction using World Bank World Development Indicators (WDI). The task is formulated as a regression problem, where macro-level indicators at year t are used to predict China's life expectancy at year $t+1$. To support this task, we built a complete workflow including data preprocessing, feature engineering, time-series dataset splitting, model training, evaluation, and frontend visualization. The dataset was constructed from China-related WDI indicators, and the prediction target was defined as life_exp_next_year. Original features, lag-1 features, and percentage change features were used to capture both current conditions and temporal changes. The project compares five models: Linear Regression, Ridge, Lasso, Random Forest, and XGBoost. A 4-fold TimeSeriesSplit strategy was used for training and validation, while the years 2019–2022 were reserved as the final test set. The results show that Lasso achieved the best performance among the baseline models, with an RMSE of 0.7993 and an MAE of 0.6951. Among the advanced models, XGBoost performed best, with an RMSE of 0.7992 and an MAE of 0.7935. In addition, a browser-based frontend visualization system was developed to support model comparison and result interpretation. Overall, this project provides an interpretable and reproducible pipeline for life expectancy prediction and demonstrates the value of combining machine learning models with interactive visualization.

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1. Introduction

1.1 Project Background and Motivation

Life expectancy is an important indicator of public health and social development. It reflects not only the general health condition of a population, but also the combined influence of economic development, healthcare resources, environmental quality, and demographic structure. Therefore, studying life expectancy prediction can provide useful insights into long-term health and social development trends.

In this project, China is selected as the study target. The World Bank World Development Indicators provide structured macro-level data covering economic, health, environmental, and demographic variables. These indicators make it possible to build a data-driven prediction model for China's next-year life expectancy. Instead of only describing historical changes, this project formulates the task as a supervised regression problem, where indicators at year t are used to predict life expectancy at year t+1.

1.2 Objectives of the Project

The main objective of this project is to predict China's next-year life expectancy using World Bank WDI indicators. To achieve this goal, the project builds a complete workflow including data preprocessing, feature engineering, time-series dataset splitting, model training, result evaluation, and frontend visualization.

Specifically, the project aims to construct a model-ready dataset, compare baseline linear models with advanced tree-based models, evaluate their prediction performance using regression metrics, and develop a browser-based frontend system to support result visualization and interpretation.

1.3 Report Organization

The rest of this report is organized as follows. Section 2 defines the prediction problem and introduces the data source. Section 3 describes the data preprocessing module, including target construction, feature engineering, and dataset splitting. Sections 4 and 5 present the baseline algorithm module and the advanced algorithm module. Section 6 introduces the evaluation metrics. Section 7 reports and analyzes the experimental results. Section 8 describes the frontend visualization system. Section 9 discusses the main findings, limitations, and practical significance. Finally, Section 10 concludes the report and presents future work.

2. Problem Definition and Data Source

2.1 Problem Definition

This project formulates China's next-year life expectancy prediction as a regression task. The model uses World Bank World Development Indicators (WDI) at year t as input features and predicts China's life expectancy at year t+1. The prediction target is defined as life_exp_next_year.

In this setting, each sample represents one year of macro-level indicators. The input variables include economic, healthcare, environmental, and demographic indicators, while the output is a continuous numerical value representing next-year life expectancy. Therefore, regression models are suitable for this task.

The prediction setting can be summarized as follows:

Table 1. Prediction Task Definition

Item	Description
Input	WDI indicators at year *t*
Output	Life expectancy at year *t+1*
Target variable	life_exp_next_year
Task type	Regression

2.2 Data Source

The dataset used in this project is built from the World Bank World Development Indicators API. The WDI database provides structured macro-level development indicators for different countries and regions. In this project, China is selected as the study target, and its country code is CHN.

The selected indicators cover several categories that may be related to life expectancy, including economic development, healthcare resources, environmental quality, and demographic structure. These indicators provide the basis for constructing the model input features.

The raw data is collected and processed into a year-based dataset. Each row corresponds to one year, and each column corresponds to a WDI indicator or an engineered feature. This structure makes the dataset suitable for time-based regression modeling.

2.3 Research Scope

The research scope of this project is summarized in Table 2.

Table 2. Research Scope of the Project

Item	Description
Country	China (CHN)
Data source	World Bank WDI API
Modeling years	1995-2022
Prediction target	Next-year life expectancy (life_exp_next_year)
Final test years	2019-2022
Task type	Regression

The modeling period covers the years from 1995 to 2022. The final test period is 2019–2022, which contains four yearly samples. Since the dataset is annual and country-level, the sample size is naturally limited. This limitation is considered in the later evaluation and discussion sections, especially when interpreting metrics such as R^2.

3. Data Preprocessing

3.1 Data Source, Sample, and Features

The raw data were obtained from the World Bank World Development Indicators (WDI) database. Ten base features were selected to represent economic conditions, health expenditure, environmental exposure, demographic structure, and health-care resources, including GDP per capita, health expenditure as a percentage of GDP, per-capita health expenditure, annual mean PM2.5 exposure, urban population share, proportion of population aged 65 years and above, unemployment rate, hospital beds per 1,000 people, infant mortality rate, and total fertility rate. These features were finally selected by referring to previous studies and relevant literature. (Wang et al., 2022; Zhao et al., 2014; Polcyn et al., 2023; Luo et al., 2021). The current life expectancy is retained as the present health status feature and further used to construct the life expectancy label for the following year. Descriptive statistics for the ten core features are shown in Table 3.

Table 3. Descriptive Statistics of Core Features

Feature	WDI Code / Source	Mean	Std.	Min	Max
Life expectancy at birth, total (years)	`SP.DYN.LE00.IN`	74.8870	2.4531	70.3630	78.2020
GDP per capita (constant 2015 US$)	`NY.GDP.PCAP.KD`	5638.1463	3342.8894	1545.1943	11830.5984
Current health expenditure (% of GDP)	`SH.XPD.CHEX.GD.ZS`	4.5565	0.5110	3.6174	5.8808
Current health expenditure per capita (current US$)	`SH.XPD.CHEX.PC.CD`	278.8399	199.1751	42.8080	755.4360
PM2.5 air pollution, mean annual exposure (micrograms per cubic meter)	`EN.ATM.PM25.MC.M3`	47.7137	5.9188	34.8105	58.4662
Urban population (% of total population)	`SP.URB.TOTL.IN.ZS`	47.9311	11.3658	29.0404	65.2176
Population ages 65 and above (% of total population)	`SP.POP.65UP.TO.ZS`	8.9437	2.1756	6.0531	13.7819
Unemployment, total (% of total labor force)	`SL.UEM.TOTL.ZS`	4.2425	0.6035	3.0000	5.0000
Hospital beds (per 1,000 people)	`SH.MED.BEDS.ZS`	3.0043	1.1882	1.6900	5.3800
Infant mortality rate (per 1,000 live births)	`SP.DYN.IMRT.IN`	17.5250	11.0533	4.8000	38.7000
Fertility rate, total (births per woman)	`SP.DYN.TFRT.IN`	1.5820	0.1836	1.0340	1.7980
Next-year life expectancy target	`Derived from SP.DYN.LE00.IN (shift -1)`	75.1581	2.3519	70.8230	78.2020

After label construction, the final modeling window spans 1995–2022, with a total of 28 annual observations. The final feature space contains 31 features: 11 contemporaneous level terms, 10 first-lag terms, and 10 relative-change terms. Among them, the contemporaneous level terms include current-year life expectancy and 10 basic explanatory features. The remaining features capture the lagged levels and year-on-year changes of these basic explanatory features. This design retains both cross-sectional level information and short-run temporal dynamics for next-year life-expectancy prediction.

3.2 Preprocessing Pipeline

The full preprocessing pipeline is illustrated in Figure 1 below.

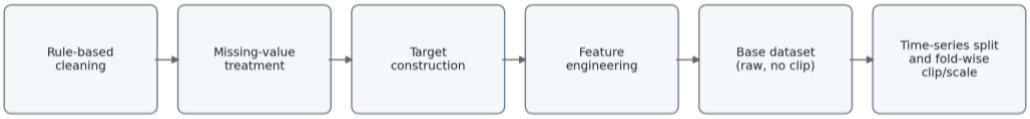

Figure 1. Preprocessing Pipeline

3.2.1 Data Cleaning and Preparation

Preprocessing removes duplicate records by year and imposes rule-based constraints on the raw features. The specific rules are as follows:

- Non-negativity constraints are applied to contemporaneous life expectancy, GDP per capita, health expenditure as a share of GDP, per capita health expenditure, PM2.5 exposure, hospital beds per 1,000 people, infant mortality rate, and total fertility rate.
- 0-100 range constraints are imposed on health expenditure as a percentage of GDP, urban population share, population aged 65 and above, and unemployment rate. Any violating observation is set to missing

Forward filling in ascending year order is then performed on contemporaneous life expectancy and the ten basic explanatory features, with any remaining missing values imputed using the full-sample median. Based on this, we construct the next-year life expectancy target, ten first-order lag features, and ten relative-change features. The relative-change term is defined as:

$$\frac{(current\ value - previous\ value)}{previous\ value}$$

and stored as a ratio.

After feature construction, forward filling and median imputation are repeated to ensure a complete base modeling matrix.

It should be noted that after project completion, full-sample median imputation was found to be not fully consistent with the goal of strictly preventing time leakage. This limitation remains to be addressed in future work.

3.2.2 Time-Series Processing and Splitting

The base data stage retains the unclipped and unstandardized modeling matrix, and all additional transformations are performed after time-series data partitioning. The test window is fixed to start in 2019, so 1995-2018 forms the train-validation segment and 2019-2022 forms the final test segment. Within the train-validation segment, four-fold time-series cross-validation with expanding training windows is adopted, yielding the following splits: 1995-2002 / 2003-2006, 1995-2006 / 2007-2010, 1995-2010 / 2011-2014, and 1995-2014 / 2015-2018. The overall framework for time-series data partitioning is illustrated in Figure 2.

For each fold, IQR clipping bounds are estimated exclusively on the training fold, where

$$Q1 - 1.5 \times IQR \quad \text{and} \quad Q3 + 1.5 \times IQR$$

serve as the lower and upper bounds and are then applied to the corresponding validation fold. Standardization is handled in the same way by fitting `StandardScaler` only on the training fold and transferring the fitted parameters to validation data. The final test-set transformations are estimated from the complete train-validation segment.

Under the current base dataset, 47 observations would be flagged as outliers if the same IQR rule were used only for detection. These outlying samples are not directly processed at the base stage. Instead, they are clipped in the subsequent fold-wise clip versions using bounds estimated from the training fold.

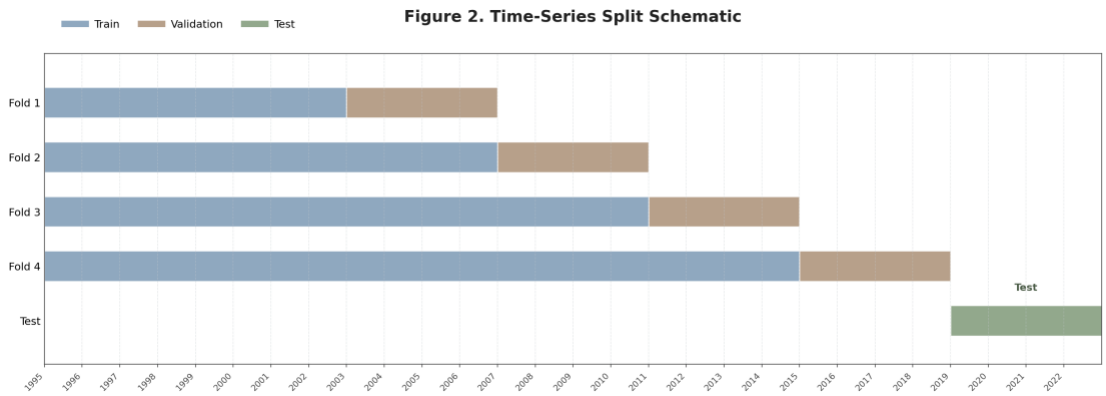

Figure 2. Time-Series Split Schematic

3.3 Summary and Preprocessing Limitations

Overall, the preprocessing stage implements rule-based cleaning of annual raw data, missing value imputation, target and dynamic feature construction, as well as time-ordered fold-wise transformation and splitting. This produces a well-structured data foundation ready for downstream model comparison. However, several issues in the data preprocessing stage were challenged during the peer review process. These issues are explained and analyzed below.

First, the train-validation stage includes only 24 observations with 31 features, resulting in a typical N < P setting. This stems primarily from two factors: the preprocessing strategy prioritizes retaining as many features as reasonably relevant to life expectancy, and the study focuses on annual forecasting for a single country, which inherently restricts the sample size. We do not extend the analysis to global life expectancy prediction, since a single-country setting is more conducive to yielding context-specific and highly interpretable findings, even at the cost of a reduced sample size. Accordingly, the four-fold results should be regarded as exploratory rather than robust statistical evidence.

In addition, the overlap between the 2019–2022 test window and the COVID-19 shock period may lead to structural shifts in the results. As mentioned earlier, full-sample median imputation still carries residual risks of temporal leakage, and more potential issues may exist. Future improvements can be made by incorporating fold-wise feature screening at the modeling stage and, if necessary, introducing a pre-COVID control window.

4. Linear Baseline Module

4.1 Baseline model principles

This project uses three linear baseline models: Ordinary Least Squares Linear Regression, Ridge Regression, and Lasso Regression. These models assume that the prediction target can be represented as a linear combination of the input features. They are included as transparent and interpretable baselines for comparison with more advanced tree-based models.

Ordinary Least Squares Linear Regression estimates coefficients by minimizing the sum of squared residuals without adding any penalty term. This makes it the simplest linear baseline, but it can become unstable when predictors are highly correlated or when the number of features is large relative to the number of training samples.

Ridge Regression extends the linear model by adding an L2 penalty to the loss function. This penalty shrinks coefficients toward zero and can reduce variance, but Ridge usually keeps all features in the model with non-zero weights.

Lasso Regression adds an L1 penalty to the loss function. In addition to shrinking coefficients, Lasso can set some coefficients exactly to zero. Therefore, it performs implicit feature selection, which is useful in this project because the dataset has a small number of yearly observations but a relatively large number of engineered features.

4.2 Implementation Details

The baseline models are implemented in baseline_models/train_baseline_models.py. The implementation uses the time-ordered train-validation folds generated by split_time_series_datasets.py. The training and validation pool covers 1995–2018, while the held-out test set covers 2019–2022.

The baseline module uses the clip_scaled data variant. In this variant, IQR clipping bounds are fitted only on the training fold and then applied to the validation fold. StandardScaler is also fitted only on the training fold and then applied to validation data. For the final test evaluation, the scaler is fitted on the full train-validation segment and applied to the test set. This design helps reduce data leakage in the time-series setting.

After excluding the year column and the target variable, the baseline models use 31 predictors. Linear Regression is trained without regularization. Ridge and Lasso use grid search over the regularization strength alpha, and the best alpha is selected by minimizing the mean validation RMSE across the four folds.

4.3 Hyperparameter Search

For Ridge and Lasso, the following alpha search space is used:

[0.001, 0.005, 0.01, 0.05, 0.1, 0.5, 1.0, 5.0, 10.0, 50.0, 100.0]

The selected best alpha values are:

Table 4. Selected Alpha Values for Ridge and Lasso

Model	Best Alpha
Ridge	0.5
Lasso	0.05

5. Advanced Algorithm Module

5.1 Advanced Models

In this stage, we used two tree-based regression models, Random Forest (RF) and XGBoost, to predict next-year life expectancy. These models were chosen because life expectancy is influenced by multiple socioeconomic, health, and demographic factors, and the relationships between these variables and the target may be nonlinear. Random Forest was included as a representative bagging-based ensemble method with interpretable feature importance, while XGBoost was used as a boosting-based tree model that performs strongly on structured tabular data.

5.2 Parameter Search and Experimental Setting

To improve model selection, we first conducted parameter search using 4-fold evaluation for both Random Forest and XGBoost. For each fold, the model was trained on the corresponding training split and evaluated on the validation split, and the average MAE and RMSE were used to compare parameter settings.

For Random Forest, we performed a small grid search over three main hyperparameters: the number of trees (n_estimators), maximum tree depth (max_depth), and minimum leaf size (min_samples_leaf). The search explored several candidate combinations, including shallow and deeper trees as well as different ensemble sizes. The final model adopted n_estimators = 100, max_depth = 5, and min_samples_leaf = 1.

For XGBoost, we tuned the number of boosting rounds, tree depth, and learning rate, while keeping subsample and colsample_bytree fixed at 0.8. The explored settings covered both smaller and larger boosting rounds and two learning-rate levels. The final

model used n_estimators = 100, max_depth = 3, learning_rate = 0.03, subsample = 0.8, and colsample_bytree = 0.8.

6. Evaluation Metrics

To evaluate the prediction performance of different models, this project uses several standard regression metrics, including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), coefficient of determination (R^2), and Mean Absolute Percentage Error (MAPE). These metrics are calculated by comparing the actual life expectancy values with the predicted values on the test set. Since the final test set only contains four yearly samples from 2019 to 2022, multiple metrics are used together to provide a more reliable evaluation.

6.1 Mean Absolute Error

Mean Absolute Error (MAE) measures the average absolute difference between the actual values and the predicted values. It directly reflects the average prediction error in the same unit as the target variable.

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|$$

where y_i is the actual value, \hat{y}_i is the predicted value, and n is the number of test samples. A lower MAE indicates better prediction accuracy.

6.2 Root Mean Squared Error

Root Mean Squared Error (RMSE) measures the square root of the average squared prediction error. Compared with MAE, RMSE gives larger penalties to large errors, so it is useful for identifying models that occasionally make large prediction mistakes.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

A lower RMSE means that the model predictions are closer to the actual values overall.

6.3 Coefficient of Determination

The coefficient of determination, R^2, measures how well the model explains the

variation in the target variable compared with a simple mean-based prediction.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

where $\bar{y}$ is the mean of the actual values. In general, a higher R^2 indicates better explanatory power. However, in this project, R^2 should be interpreted carefully because the final test set contains only four samples and the life expectancy values vary within a narrow range. Under this condition, even small prediction errors can lead to very negative R^2 values.

6.4 Mean Absolute Percentage Error

Mean Absolute Percentage Error (MAPE) measures the average absolute prediction error as a percentage of the actual value. It is useful because it provides an intuitive relative error measure.

$$MAPE = \frac{100\%}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right|$$

A lower MAPE indicates better prediction performance. Since life expectancy values are positive, MAPE is suitable for this task and provides an easy-to-understand percentage-based evaluation.

6.5 Metric Interpretation in This Project

In this project, MAE, RMSE, and MAPE are the main metrics used to compare model performance. R^2 is also reported, but it is treated as a supplementary metric because the final test set contains only four samples. A negative R^2 means that the model performs worse than a simple mean-based predictor in terms of explained variance. However, the life expectancy values in the test period vary within a very narrow range, so R^2 becomes highly sensitive to small prediction errors. Therefore, a negative R^2 does not mean that the model comparison is meaningless. Instead, this project mainly uses MAE, RMSE, and MAPE to compare the absolute and relative prediction errors of different models. Under this interpretation, Lasso can still be considered the best baseline model because it achieves the lowest RMSE, MAE, and MAPE among the baseline models.

7. Experimental Results and Analysis

This section presents the experimental results of the baseline models and advanced models. The baseline models include Linear Regression, Ridge Regression, and Lasso Regression, while the advanced models include Random Forest and XGBoost. The evaluation is mainly based on MAE, RMSE, and MAPE, while R^2 is reported as a supplementary metric because the final test set contains only four samples.

7.1 Baseline Model Results

The baseline model results are shown in Table 5. Among the three baseline models, Lasso achieved the best performance on the final test set. It obtained the lowest RMSE, MAE, and MAPE values, indicating better prediction accuracy compared with Linear Regression and Ridge Regression.

Table 5. Baseline Model Test Results

Metric	Linear Regression	Ridge	Lasso
RMSE	1.5016	2.0398	0.7993
MAE	1.1845	1.8475	0.6951
R^2	-250.30	-462.68	-70.21
MAPE(%)	1.52	2.37	0.89

The results show that Lasso clearly outperformed the other two baseline models. This is mainly because Lasso uses L1 regularization, which can shrink some coefficients to zero and perform implicit feature selection. In this project, the dataset contains a relatively small number of yearly samples but multiple engineered features, so feature selection is useful for reducing overfitting and improving generalization.

Linear Regression performed worse because it does not include any regularization. In a small-sample setting with many features, it is more likely to fit noise in the training data. Ridge Regression applies L2 regularization and can shrink coefficients, but it keeps all features in the model. As a result, Ridge did not perform as well as Lasso in this experiment.

The R^2 values of all three baseline models are negative. This is mainly due to the very small final test set and the narrow range of life expectancy values during 2019–2022. Therefore, R^2 is not the most reliable metric in this project, and MAE, RMSE, and MAPE are more useful for interpreting model performance.

7.2 Advanced Model Results

The advanced model results are shown in Table 6. The two advanced models used in this project are Random Forest and XGBoost. Both models are tree-based methods that can capture nonlinear relationships between input indicators and the prediction target.

Table 6. Test Performance of Advanced Models

Model	RMSE	MAE	R^2	MAPE(%)
Random Forest (no_clip)	1.0398	1.0266	-119.4944	0.0131
XGBoost (no_clip)	0.7992	0.7935	-70.1732	0.0102

XGBoost achieved lower RMSE and MAE than Random Forest on the final test set. This suggests that XGBoost captured nonlinear relationships more effectively under the current experimental setting. Unlike Random Forest, which averages predictions from multiple independent trees, XGBoost builds trees sequentially and attempts to correct the errors made by previous trees. This boosting mechanism may explain why XGBoost performed better in this task.

Random Forest still produced reasonable results, but its prediction accuracy was weaker than XGBoost. Since the dataset is small, the advantage of Random Forest may be limited. In contrast, XGBoost can better use sequential error correction to improve prediction performance.

7.3 Overall Model Comparison

Overall, Lasso is the best baseline model, while XGBoost is the best advanced model. Lasso and XGBoost are the two strongest models in the current experiment. XGBoost achieved the lowest RMSE, while Lasso achieved the lowest MAE and MAPE among the models with MAPE reported.

This comparison also shows that more complex models do not always completely dominate simpler models, especially when the dataset is small. Lasso remains highly competitive because it reduces model complexity and performs feature selection. XGBoost also performs well because it can capture nonlinear patterns and feature interactions. Therefore, both interpretability and nonlinear modeling ability are important in this prediction task.

7.4 Result Interpretation

The experimental results provide several important findings. First, regularization is important for the baseline models. Linear Regression performed worse because it has no regularization and is more sensitive to noise and multicollinearity in a small-sample, high-dimensional setting. Ridge Regression applies L2 regularization, but it still keeps all features in the model, so redundant or highly correlated indicators may continue to affect prediction. In contrast, Lasso achieved the best baseline performance through L1 regularization and feature selection. This suggests that reducing redundant features is helpful when the number of observations is limited.

Second, the advanced model results show that XGBoost performed better than Random Forest on the final test set. This suggests that boosting-based tree learning was more effective than bagging-based learning under the current setting. However, because the final test set contains only four samples, the difference between the two advanced models should be interpreted carefully.

The feature importance results provide additional insight into how the two advanced models make predictions. For Random Forest, feature importance is distributed across several health, economic, and demographic indicators, including infant mortality, current life expectancy, GDP per capita, population aged 65 and above, and urban population share. This suggests that Random Forest uses a broader group of variables when making predictions.

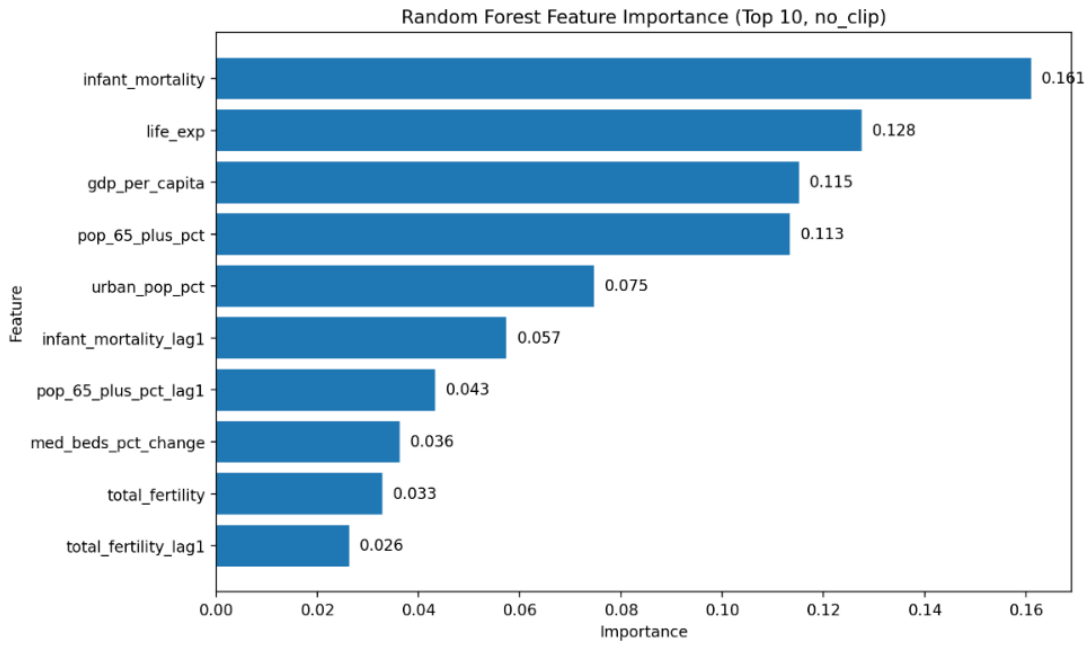

Figure 3. Random Forest Feature Importance

In contrast, XGBoost relies more heavily on a few high-level variables, especially urban population share, current life expectancy, and GDP per capita. This concentrated feature usage may help explain why XGBoost produced more stable predictions across the four test years.

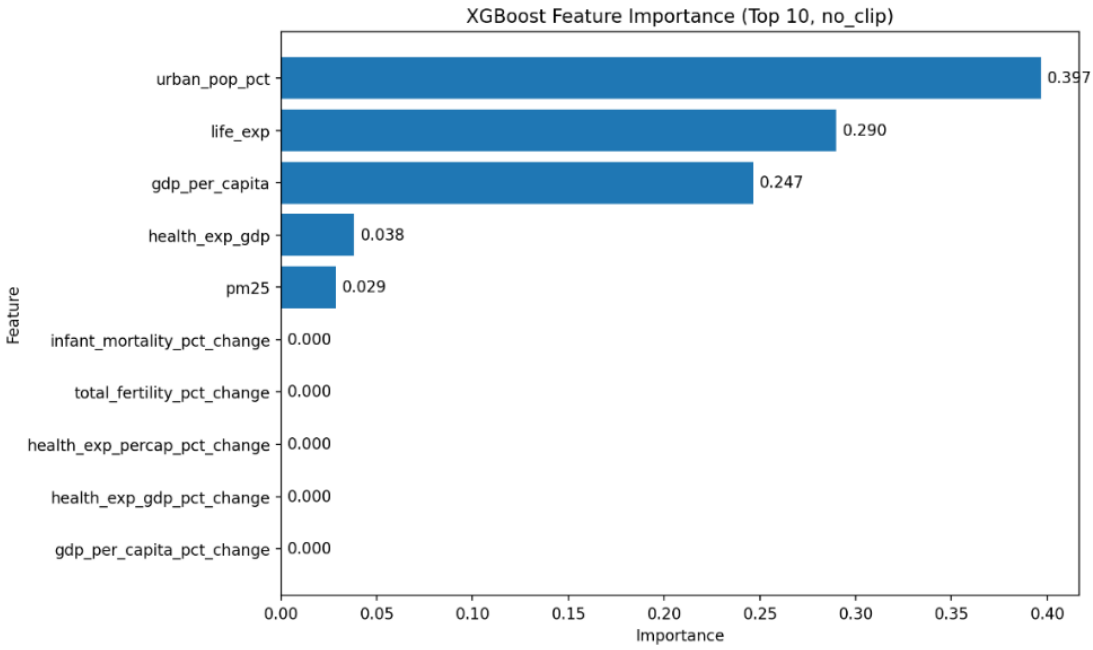

Figure 4. XGBoost Feature Importance

The prediction curves further show the difference between the two advanced models. Random Forest produced slightly different predictions across the test years, but its predictions were consistently lower than the actual values. This indicates a downward bias in the Random Forest predictions.

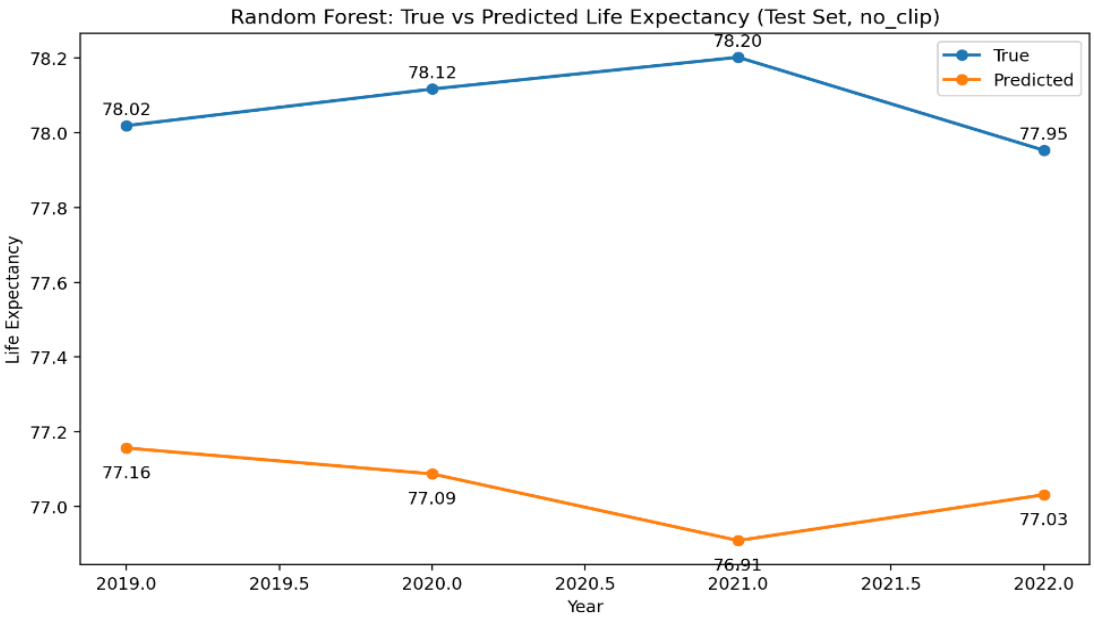

Figure 5. Random Forest True vs Predicted Values

XGBoost generated almost identical predictions for the four test years. This indicates that it mainly learned the general level of life expectancy rather than detailed year-to-year variation. Therefore, XGBoost performed better mainly because it achieved smaller overall deviation, not necessarily because it captured yearly changes more accurately.

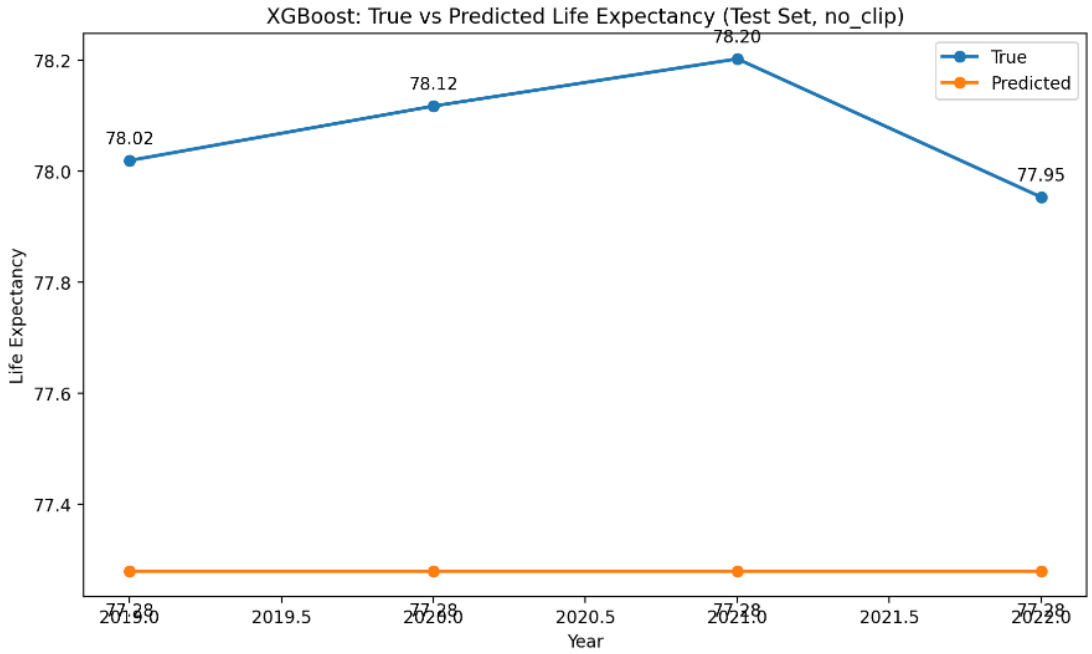

Figure 6. XGBoost True vs Predicted Values

The comparison between Lasso and XGBoost is also important. XGBoost had a slightly lower RMSE, while Lasso had a lower MAE. This means that XGBoost may reduce larger errors slightly better, while Lasso produces lower average absolute error. Considering the small test set, both models should be regarded as competitive models rather than declaring one as absolutely superior.

Finally, the evaluation metrics and interpretation results should be understood carefully. Since the test target values are very close to each other, even small prediction errors can lead to large negative R^2 values. Therefore, this project mainly relies on MAE, RMSE, and MAPE for model comparison. In addition, feature coefficients and feature importance results only indicate predictive associations. Tree-based feature importance does not show whether a feature has a positive or negative effect, and neither coefficients nor feature importance should be interpreted as causal evidence.

8. Frontend Visualization and Result Presentation

This section introduces the frontend visualization system developed for this project. In addition to generating model outputs and result files, the project provides a browser-based interface to make the results easier to compare, interpret, and present.

8.1 Frontend System Design

The frontend system is implemented as a standalone HTML page. It reads the prediction results, model metrics, coefficients, and feature importance files generated by the modeling modules. Users can open the page through a local HTTP server and switch between different models directly in the browser.

The main purpose of the frontend is to organize model results into a clear visual dashboard instead of only presenting separate CSV, JSON, or image files. This makes the final results more accessible and easier to understand.

8.2 Supported Functions

The frontend supports several core functions:

- switching between different models;
- displaying evaluation metrics such as RMSE, MAE, R^2, and MAPE;
- showing Actual vs Predicted charts;
- showing Coefficients for linear models or Feature Importance for tree-based models;
- showing Residuals on the test set.

The supported models include Linear Regression, Ridge, Lasso, Random Forest, and XGBoost.

8.3 Visualization Analysis

The frontend visualizations help explain model performance from different perspectives. The Actual vs Predicted chart shows how closely the predicted values follow the true life expectancy values. The Coefficients or Feature Importance chart helps identify which variables are more important for prediction. The Residual plot shows the prediction error for each test year.

Overall, the frontend improves the presentation quality of the project and helps users compare models more efficiently.

9. Discussion

9.1 Main Findings

The experimental results show several important findings. First, regularization is important in this small-sample prediction task. Among the baseline models, Lasso achieved the best performance because its L1 regularization can shrink some coefficients to zero and reduce feature redundancy. This is especially useful when the number of features is relatively large compared with the number of annual observations.

Second, XGBoost performed better than Random Forest among the advanced models. This suggests that boosting-based tree learning was more effective than bagging-based learning under the current setting. By building trees sequentially and correcting previous errors, XGBoost was able to control prediction error more effectively on the final test set.

Third, Lasso and XGBoost were the two strongest models in the overall comparison. XGBoost achieved slightly lower RMSE, while Lasso achieved lower MAE. This indicates that the nonlinear model may reduce larger errors slightly better, while the regularized linear model provides strong average accuracy and better interpretability. Therefore, both models are valuable in this task, and it would be inappropriate to claim that one model is absolutely superior based on such a small test set.

9.2 Limitations

Although the project completed a full pipeline from data preprocessing to frontend visualization, several limitations remain. First, the dataset is based on annual data for a single country, so the number of available samples is naturally limited. The train-validation period contains only 24 observations, while the model uses 31 features. This creates a high-dimensional small-sample setting and increases the risk of overfitting. Therefore, the current results should be interpreted as exploratory rather than statistically conclusive.

Second, the final test set contains only four samples from 2019 to 2022. As a result, evaluation metrics can be sensitive to small prediction errors. This is especially true for R^2, because the actual life expectancy values in the test period vary within a very narrow range. Therefore, MAE, RMSE, and MAPE are more reliable than R^2 for interpreting the results.

Third, the test period overlaps with the COVID-19 period, which may introduce structural breaks or concept drift. Since the models are mainly trained on pre-2019 historical patterns, the relationship between macro-level indicators and life expectancy

may change during the test period. This limits the practical certainty of the prediction results.

Fourth, the study focuses only on China. Although this makes the analysis more context-specific, it also limits the generalizability of the findings. Models trained on China's historical indicators may not perform the same way for other countries with different economic, healthcare, environmental, and demographic patterns.

Finally, the model outputs should be interpreted as predictive associations rather than causal evidence. Feature coefficients and feature importance can help identify variables that are useful for prediction, but they do not prove that these variables directly cause changes in life expectancy. In particular, tree-based feature importance does not show the direction of influence, so further interpretation methods such as SHAP or partial dependence analysis would be needed in future work.

9.3 Practical Significance

This project has practical significance because life expectancy is an important indicator of public health and social development. By using WDI indicators to predict next-year life expectancy, the project explores how macro-level indicators can be connected with health outcomes through data-driven methods.

The project also provides a reusable workflow that can be extended in future studies. The pipeline includes data collection, preprocessing, feature engineering, time-series splitting, model training, evaluation, and visualization. Therefore, it is not only a single prediction experiment, but also a framework that can be adapted to other countries or related development indicators.

The frontend visualization system further improves the usability of the project. It allows users to compare models, view prediction results, and interpret feature importance in one interface. This makes the results easier to understand, present, and further analyze.

10. Conclusion and Future Work

This project built a complete pipeline for China's next-year life expectancy prediction using World Bank WDI indicators. The workflow covers data preprocessing, feature engineering, time-series dataset splitting, baseline and advanced model training, performance evaluation, and frontend visualization. By using indicators at year t to predict life expectancy at year t+1, the project formulated the task as a regression problem and provided a systematic way to compare different machine learning models.

The experimental results show that Lasso performed best among the baseline models, while XGBoost achieved the strongest performance among the advanced models. Lasso benefited from L1 regularization and feature selection, which helped reduce overfitting in the small-sample setting. XGBoost showed stronger nonlinear modeling ability than Random Forest. However, because the dataset is small and the final test period contains only four years, these results should be interpreted as exploratory rather than conclusive. Overall, Lasso and XGBoost were the two most competitive models in this experiment, showing the value of both interpretable regularized linear models and flexible tree-based models.

In addition, the frontend visualization system improved the presentation and interpretation of the results. It allows users to switch between models and view evaluation metrics, prediction curves, feature coefficients or feature importance, and residual plots in one interface. This makes the project more complete and easier to understand.

In future work, the study can be extended to more countries to evaluate whether the learned patterns can generalize across different economic, healthcare, and demographic contexts. More advanced forecasting models and stronger hyperparameter tuning strategies can also be explored. Since the test period overlaps with the COVID-19 period, future experiments may compare different test windows or introduce a pre-COVID control window to examine the effect of structural changes. In addition, feature selection and model interpretation can be further improved by using methods such as SHAP or partial dependence analysis. The frontend system can also be enhanced with richer interactive functions to support deeper model comparison and result interpretation.

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Appendix A. Repository Structure

This appendix summarizes the main repository structure of the project. The repository contains data preprocessing scripts, baseline model scripts, advanced model scripts, output files, and the frontend visualization page.

```
CS5483-Group33-Project/
├── data_preprocessing/
│   ├── dataset/
│   └── preprocessing_scripts/
├── baseline_models/
│   ├── results/
│   └── train_baseline_models.py
├── modeling/
│   ├── outputs/
│   ├── train_rf_models.py
│   ├── train_xgboost_models.py
│   ├── final_rf_test.py
│   └── final_xgboost_test.py
├── frontend.html
├── README.md
├── Report.pdf
└── requirements.txt
```

Appendix B. Reproducibility Commands

This appendix provides the main commands used to reproduce the preprocessing, model training, evaluation, and frontend visualization steps.

```
pip install -r requirements.txt

python data_preprocessing/preprocessing_scripts/preprocess_china_lifeexp.py

python data_preprocessing/preprocessing_scripts/split_time_series_datasets.py --test-start-year 2019 --n-splits 4

python baseline_models/train_baseline_models.py

python modeling/train_rf_models.py

python modeling/train_xgboost_models.py

python modeling/final_rf_test.py

python modeling/final_xgboost_test.py

python3 -m http.server 8000
```

Appendix C. Team Contribution

Member	Main Responsibility	Detailed Contribution
Muyu LI(25%)	Data preprocessing	Collected WDI data, cleaned raw indicators, constructed `life_exp_next_year`, generated lag and percentage-change features, and prepared time-series splits.
Wendi LI(25%)	Baseline models	Implemented Linear Regression, Ridge, and Lasso models, performed alpha search, evaluated baseline results, and generated coefficient/residual outputs.
Wang CHEN(25%)	Advanced models	Implemented Random Forest and XGBoost, conducted parameter search, evaluated advanced model performance, and analyzed feature importance.
Yanglai HU(25%)	Frontend visualization and result analysis	Built the browser-based visualization page, integrated model metrics and charts, compared results, and supported report/presentation preparation.

All members participated in project discussion, result review, presentation preparation, and final report revision.